

SHADRINSK, Russian Federation

WMO: 285520

Lat: 56.067N

Lon: 63.650E

Elev: 292

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-18.7	-29.6	1.1	-23.5	-24.2	1.5	-18.5	18.2	23.7	16.1	19.4	2.3	270	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	87.6	67.8	84.2	66.5	80.9	65.3	70.9	83.1	68.9	80.1	67.1	77.5	6.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	99.2	75.9	64.9	93.2	74.2	63.1	87.5	72.5	34.8	83.1	33.3	80.4	31.8	78.1	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.8	13.1	-28.8	92.3	6.2	3.9	-33.2	95.1	-36.8	97.4	-40.3	99.6	-44.8	102.4		
(5)			-29.1	73.8	6.3	2.6	-33.6	75.7	-37.3	77.2	-40.8	78.7	-45.4	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	5.3	8.8	23.3	41.1	55.0	63.6	66.1	62.6	51.9	39.4	22.0	10.7
	DBStd	24.03	12.73	12.81	9.34	9.70	9.29	8.17	6.99	7.49	8.49	8.83	13.15	13.29
	HDD50	6206	1387	1154	827	297	56	6	0	3	76	343	839	1218
	HDD65	10262	1852	1574	1292	718	329	121	73	138	401	792	1289	1683
	CDD50	1694	0	0	0	29	212	413	500	392	133	15	0	0
	CDD65	278	0	0	0	0	20	78	108	63	9	0	0	0
	CDH74	2946	0	0	0	12	300	850	1097	589	98	0	0	0
	CDH80	899	0	0	0	1	69	278	350	180	21	0	0	0
	WSAvg	5.5	4.9	5.2	6.1	6.6	6.4	5.4	4.7	4.7	5.3	6.1	5.7	5.5
	PrecAvg	17.0	0.9	0.7	0.6	0.9	1.5	2.3	2.8	2.3	1.7	1.4	1.1	1.0
(15) Precipitation	PrecMax	24.3	2.6	2.3	1.9	2.3	4.2	4.7	5.6	4.6	6.3	3.8	2.2	2.1
	PrecMin	9.9	0.1	0.0	0.0	0.0	0.3	0.5	0.3	0.4	0.3	0.3	0.0	0.0
	PrecStd	3.3	0.6	0.5	0.4	0.6	0.9	1.2	1.3	1.1	1.4	0.9	0.5	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	38.1	47.5	74.6	86.3	90.9	92.7	90.5	83.1	67.6	47.9	37.3
		MCWB	32.9	34.0	40.3	55.4	62.0	68.2	69.8	69.4	62.1	53.2	43.0	34.4
	2%	DB	30.9	34.5	41.4	67.1	81.2	86.9	87.2	84.6	76.2	61.9	43.5	34.7
		MCWB	28.9	31.5	36.1	52.1	60.4	66.9	69.2	67.4	60.6	50.9	40.2	32.2
	5%	DB	27.3	30.8	38.5	61.5	77.0	83.1	83.7	80.4	70.3	56.6	40.2	31.8
		MCWB	25.6	28.4	34.3	49.3	59.5	66.2	67.8	66.2	58.7	48.3	37.3	29.8
	10%	DB	23.5	26.9	36.0	56.5	72.5	78.9	80.3	76.1	65.8	52.2	37.1	28.3
		MCWB	22.0	24.6	32.5	46.9	56.7	64.4	66.6	64.6	56.2	46.1	34.5	26.6
	0.4%	WB	33.4	34.9	40.8	57.0	66.1	72.5	73.0	72.3	65.2	55.2	43.6	34.8
		MCDB	35.2	37.5	46.3	73.5	77.3	85.5	84.4	84.9	76.5	64.1	46.8	36.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.0	31.5	36.9	53.2	62.3	69.6	71.2	69.7	61.7	51.9	40.4	32.6
		MCDB	30.7	34.0	40.6	64.7	76.9	81.2	83.5	81.8	73.2	60.1	43.2	34.5
	5%	WB	25.7	28.4	35.0	50.4	60.2	67.4	69.4	67.3	59.2	49.0	37.6	29.9
		MCDB	27.2	30.8	38.1	60.6	74.2	79.3	80.6	77.2	68.9	55.1	40.0	31.6
	10%	WB	22.1	24.6	32.8	47.4	58.0	65.5	67.8	65.4	56.7	46.5	34.8	26.6
		MCDB	23.6	26.7	35.8	55.8	70.5	76.7	77.9	74.3	64.6	51.9	36.9	28.5
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	15.7	15.5	17.1	20.9	19.4	18.9	18.0	16.8	12.8	10.7	11.8
		MCDBR	12.7	13.2	14.8	25.0	28.0	25.7	25.2	24.7	24.1	21.0	12.0	11.7
		MCWBR	11.9	11.4	11.0	13.3	12.7	10.6	9.6	10.3	11.6	12.2	9.2	10.2
	5% WB	MCDBR	12.4	13.0	13.4	23.0	25.2	22.6	21.9	21.9	22.3	19.5	10.5	11.8
		MCWBR	11.8	11.4	10.4	12.9	12.6	10.7	9.9	10.0	11.4	12.2	8.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.287	0.318	0.371	0.415	0.416	0.410	0.430	0.444	0.387	0.356	0.313	0.287
	taud		2.048	2.025	1.948	2.089	2.200	2.290	2.231	2.158	2.281	2.238	2.131	2.016
	Ebn at Noon		183	226	241	252	261	265	255	241	240	213	172	147
	Edh at Noon		20	31	43	44	42	39	40	41	31	25	19	16
(44) All-Sky Solar Radiation	RadAvg		253	585	1078	1397	1669	1824	1752	1358	894	454	226	156
	RadStd		21	30	63	136	158	174	140	139	78	72	29	20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page